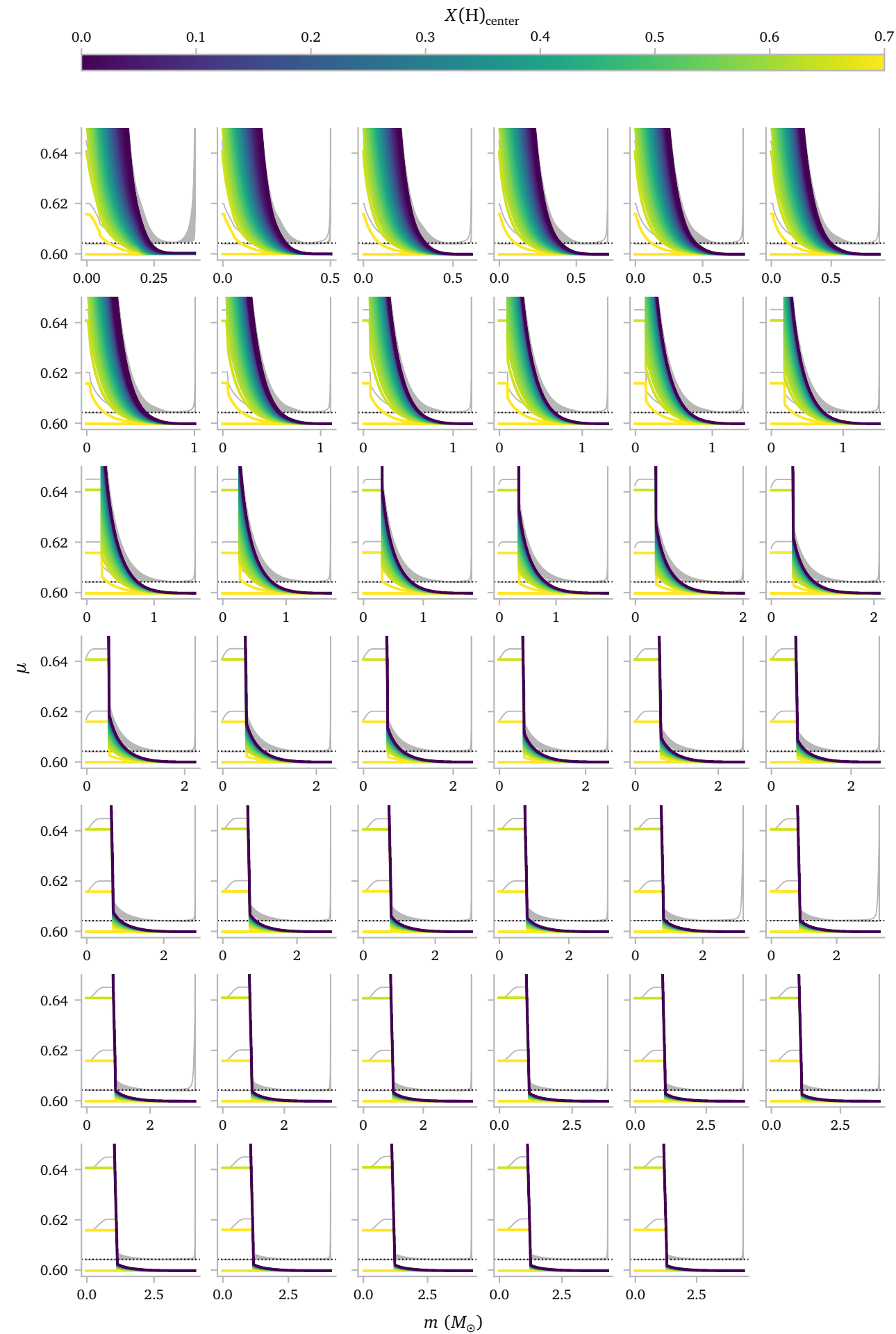


Notes Week 31

a. Fully Ionized Mean Molecular Weight

a.1 Main Sequence

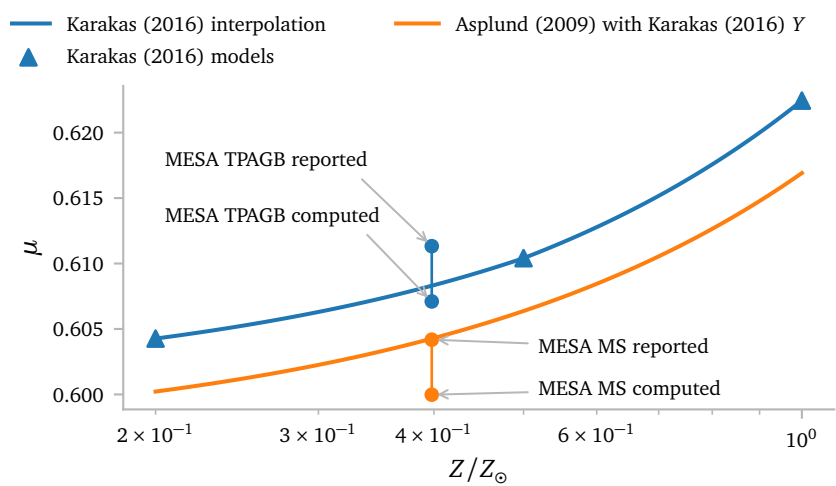
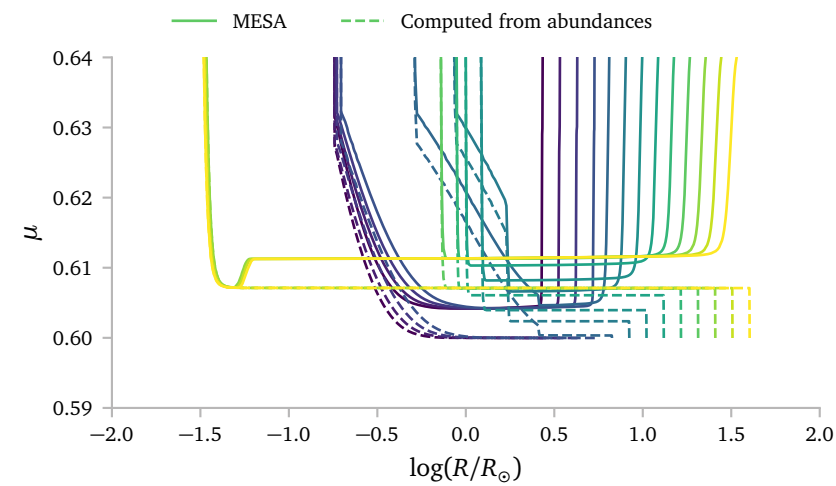
Since we are using the fully ionized mean molecular weights for the accreted material, it is only fair to compare this with the fully ionized mean molecular weights of MESA abundances.



The gray lines are the μ values as reported by *MESA*. The colored lines are the computed values from the abundances and the horizontal line is μ from [Asplund et al. \(2009\)](#). I *really* do not understand why the computed mean molecular weights from the *MESA* models are lower than reported μ from *MESA* and the computed [Asplund et al. \(2009\)](#) μ values.

I feel like the μ values reported from *MESA* are more representative because they agree with [Asplund et al. \(2009\)](#)(?)

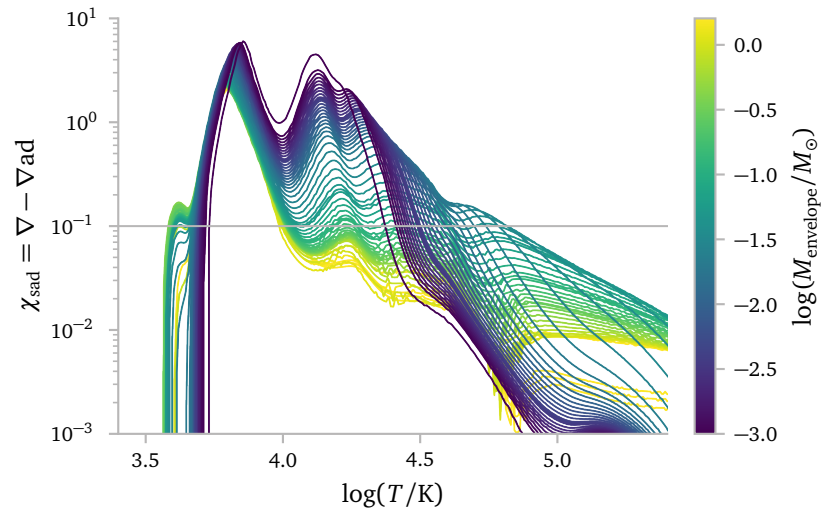
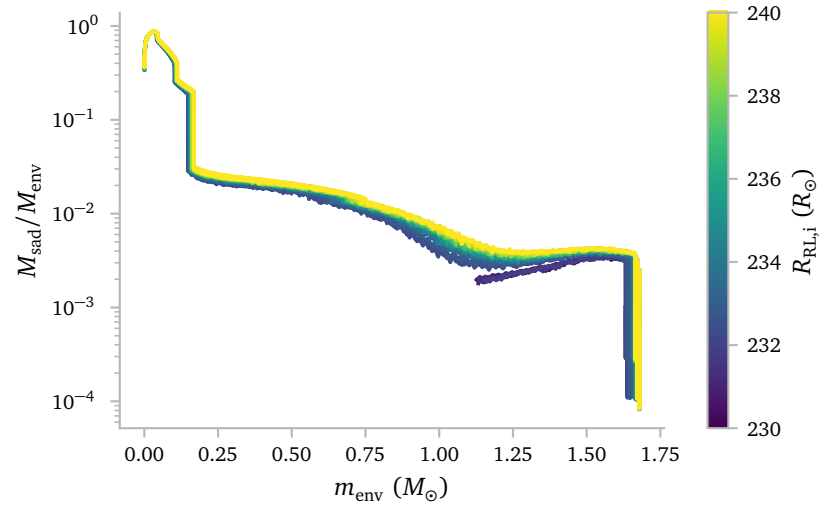
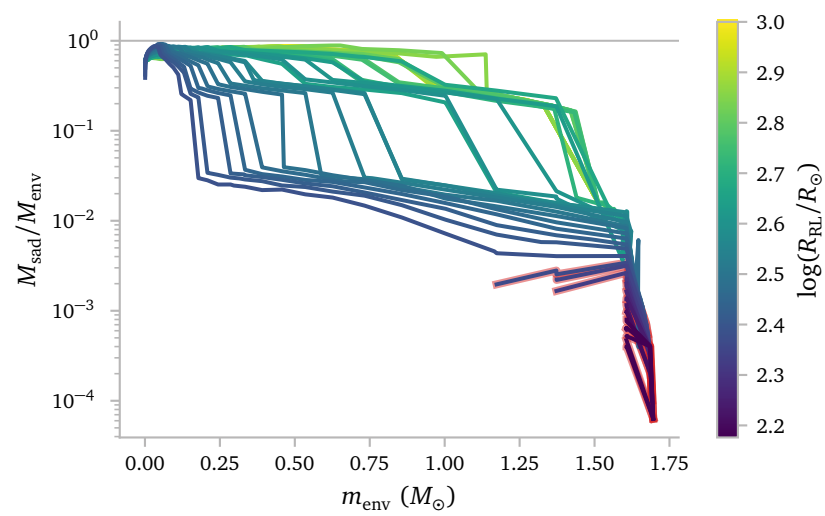
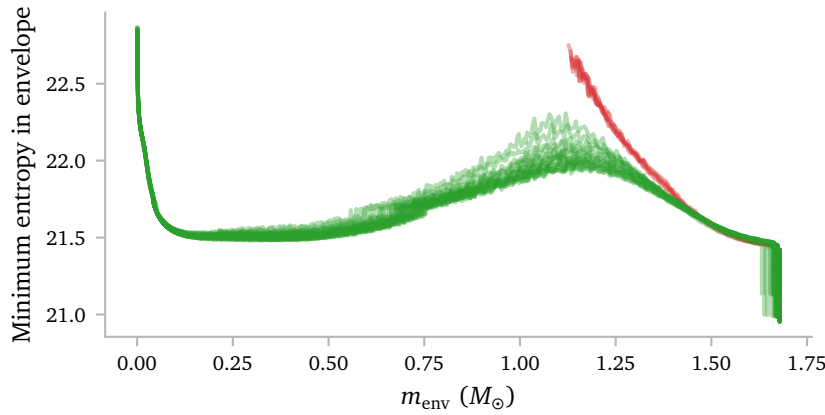
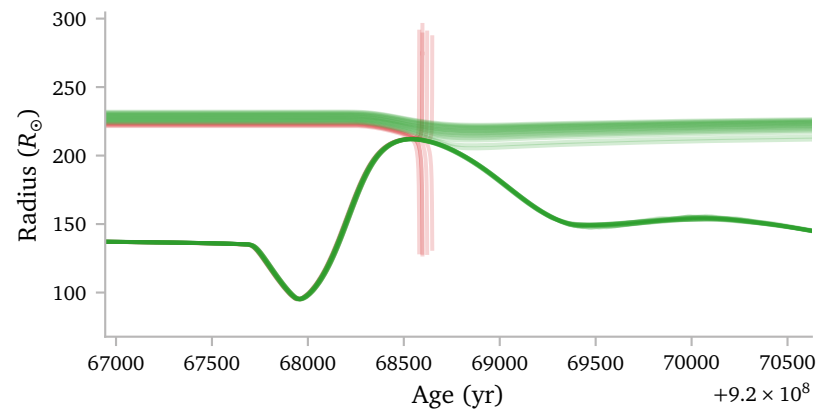
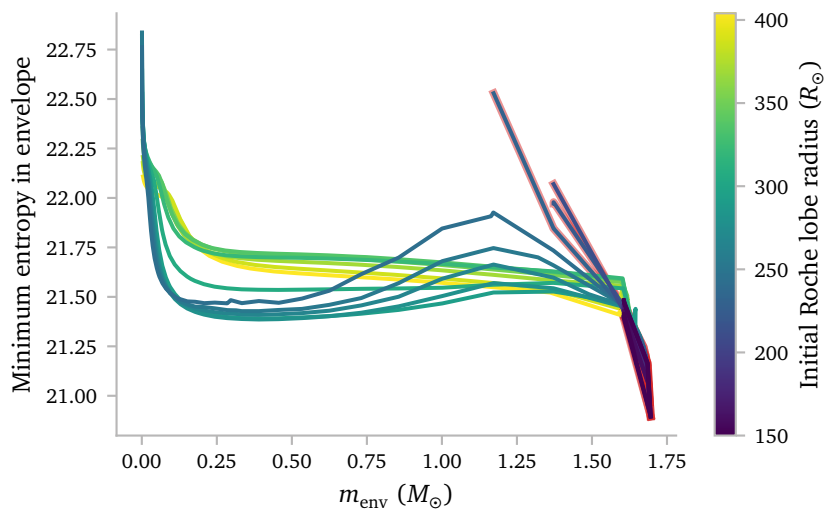
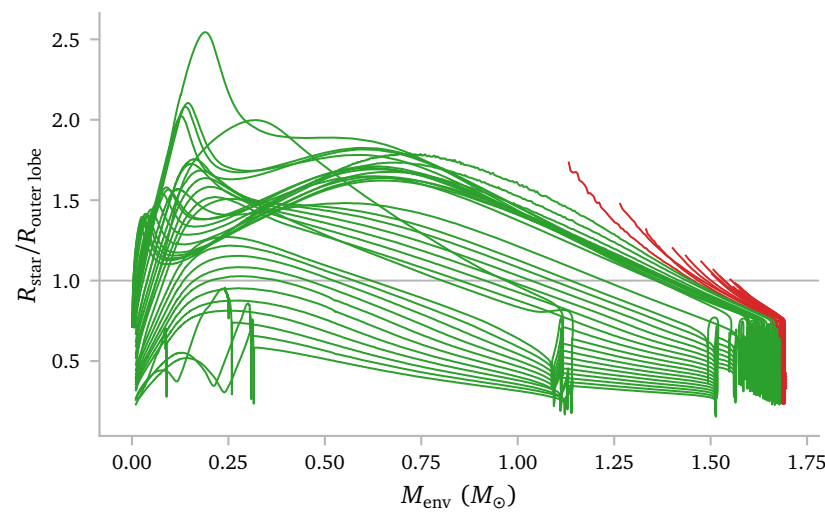
a.2 Giant branch and Thermally Pulsing Asymptotic Giant Branch



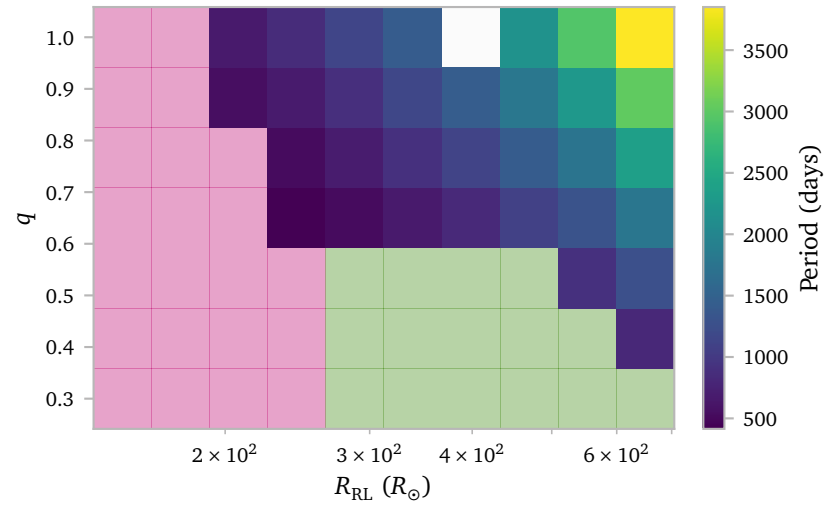
The mean molecular weights reported by the *MESA* models on the main sequence agree very well with the mean molecular weights of the [Asplund et al. \(2009\)](#).

On the TPAGB, it is exactly opposite. Here, the computed values from the abundances agrees much better with the μ computed from the [Karakas & Lugaro \(2016\)](#) abundances.

b. Failing models II



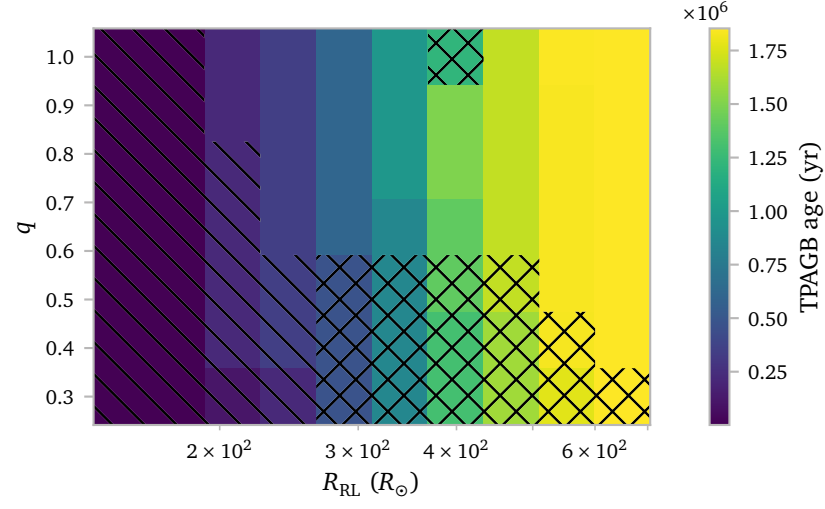
I want to return to this plot I made a while ago:



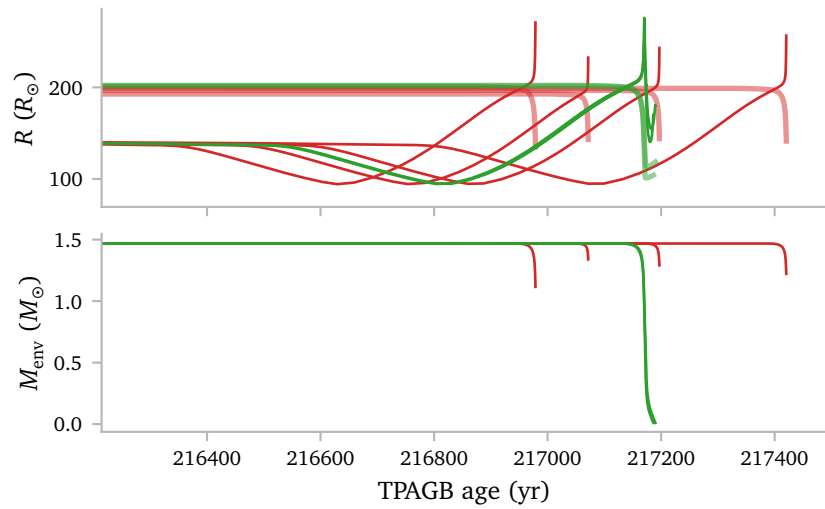
The large jumps towards lower envelope masses are artificial and originate only from the wavy nature of the superadiabicity profile. I show this in the plot below.

We do see though that the superadiabatic region is smallest in the models that crash. From the plot above, it also looks like the relative width also increases quite a lot with the age of the TPAGB star, or the initial Roche lobe radius, which is very similar in this case.

I am curious if the models that fail, all happen in during the same thermal pulses. If however, it is possible to survive RLOF during a thermal pulse with a certain q , but not with another q .

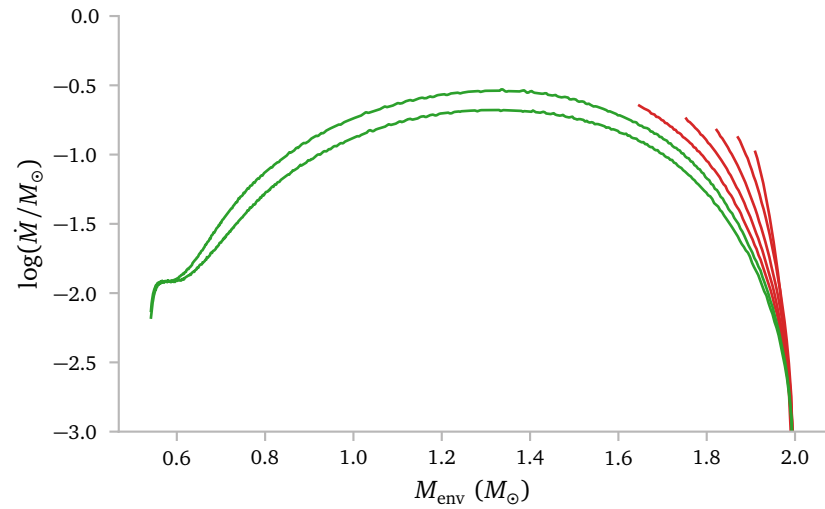


Clearly there are models that interact at roughly the same time (same color), but some of them crash, (dashed) while others do not.

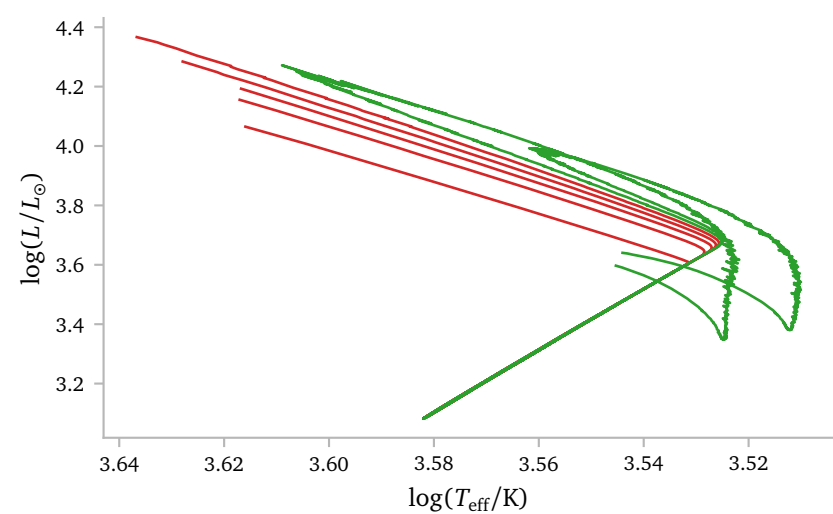


Here, I explicitly show this. The models interact during the same pulse, however, only the green lines, with the largest q , survive.

I guess the mass loss rate is another important consideration.



I show that here. The models with the largest q have the smallest $\dot{M}$ and thus strip the star slower. The differences here might seem minor, but this is a log scale so differences are actually quite large.

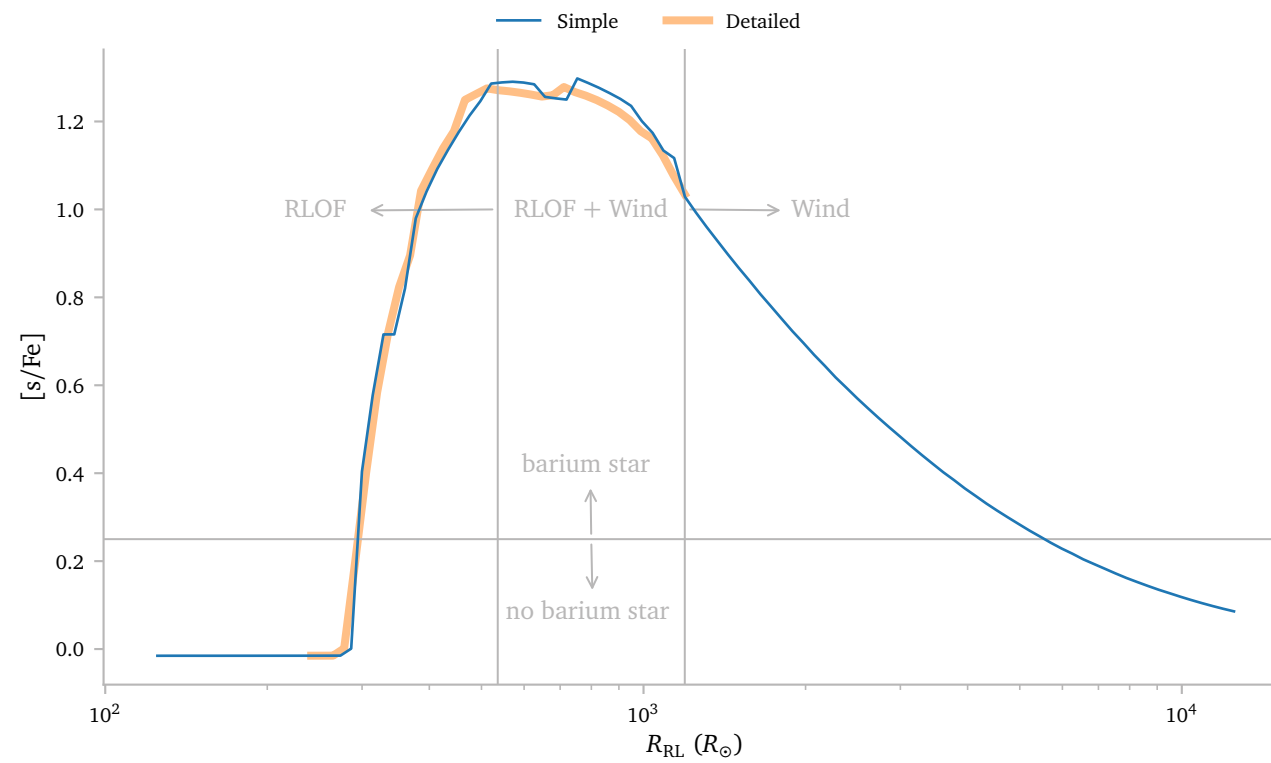


c. Grids

Using a very simple assumption, it is possible to quite accurately estimate the final accretor abundances of the detailed *MESA* simulations using only the simple direct binary evolution in python with *evolve.py*. The idea is this:

Follow the simple binary evolution as predicted by *evolve.py* exactly up to the point of RLOF. Once RLOF happens, assume that *all* of the remaining envelope mass is expelled instantly and accreted with a certain efficiency. To match *MESA* models, we should let ϵ_{MT} be this efficiency parameter. We then say that this material is accreted instantly at the last simple binary evolution step.

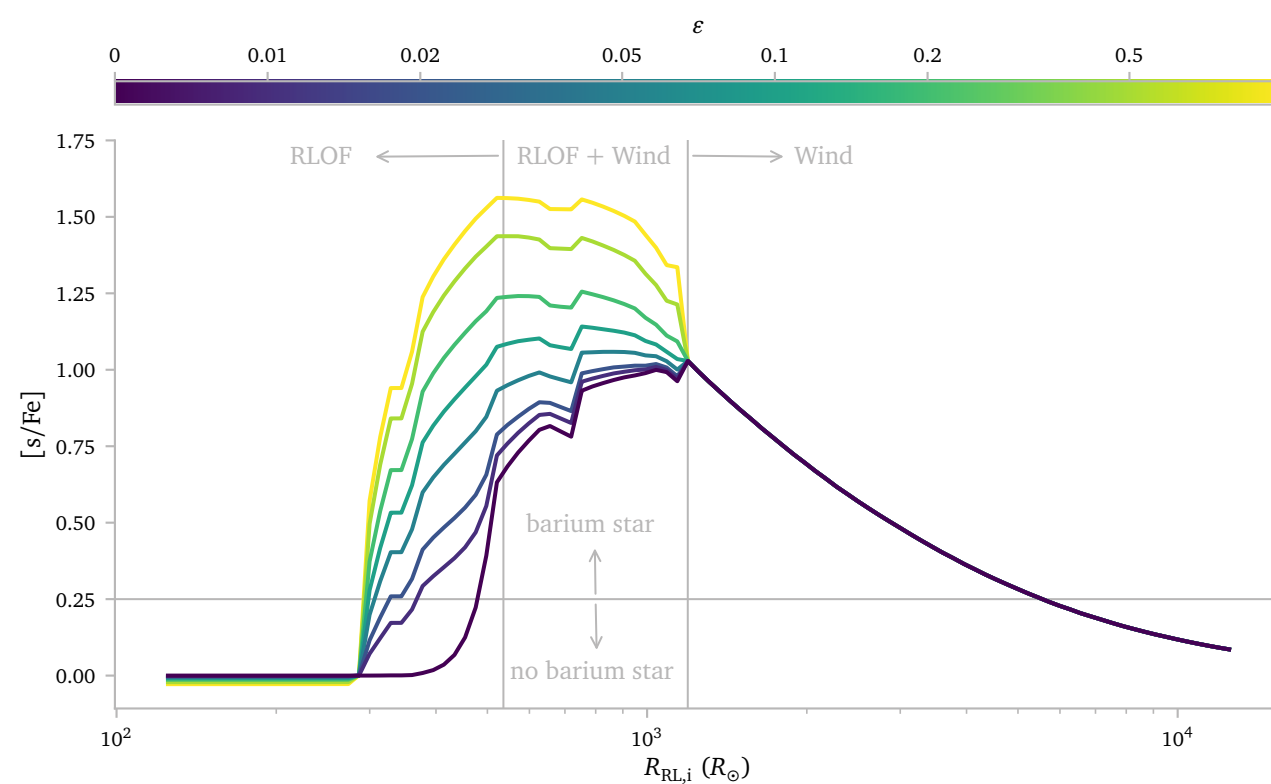
Below, I show this assumption and compare it to the R_{RL} grid that I simulated last week. These all have $M_{\text{TPGB},i} = 2.2 M_{\odot}$, $q = 0.6$ and $\epsilon = 0.25$.



Clearly, the two match very well, and the small differences are probably due to the final assumption that there is *no* wind during RLOF and the small differences between the orbital evolution of the detailed *MESA* models and the simple binary models.

It is also very easy to simulate large binaries that miss RLOF:

Below, I show the effects of the mass transfer efficiency on the final abundances.



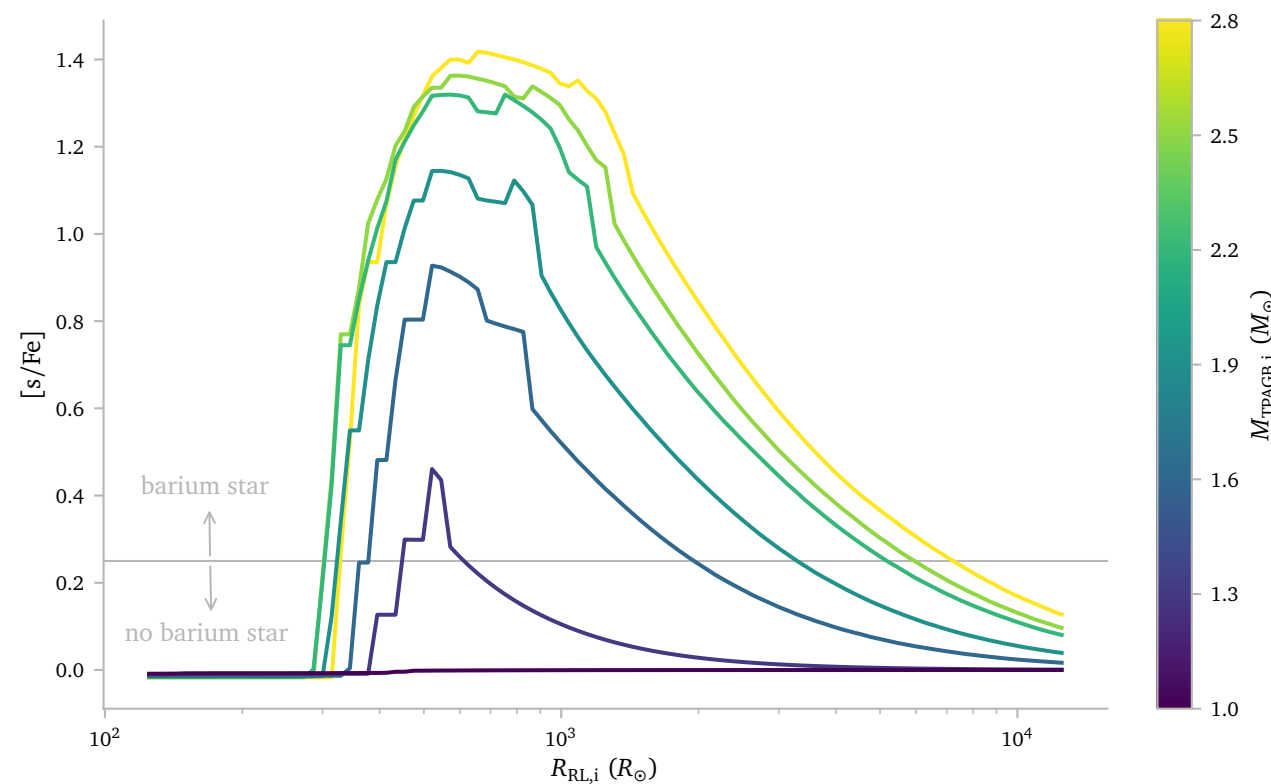
Interestingly, it seems like it is very easy to create barium stars even *without* any RLOF accretion. The enrichment for the purple line comes only from the wind accretion.

If we do have *RLOF* accretion, we can get higher abundances for smaller binaries. These binaries will also have smaller final periods. It also seems like we can create barium stars in very large binaries, just using wind accretion.

Lastly, we see a quite sharp drop between the models that undergo RLOF and those that do not. I *think* this is due to the envelope mass remaining during the last thermal pulse. If the star has, say $0.5 M_{\odot}$ of envelope remaining, this will be accreted with RLOF for models that *just* undergo RLOF, while this will *not* be accreted with the same efficiency for models that just barely miss RLOF.

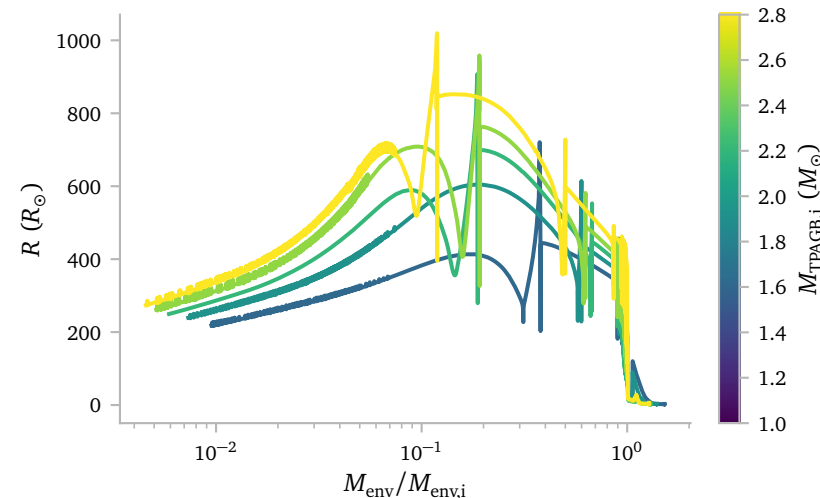
To test this, we can look at different masses, since this will shift the amount of envelope mass remaining during the last thermal pulse.

Here again, we have $q = 0.6$ and $\epsilon = 0.25$.



Here, it looks like the $1.6 M_{\odot}$ model has much more mass during the last thermal pulse than for example, the $2.8 M_{\odot}$ star.

I confirm this below.

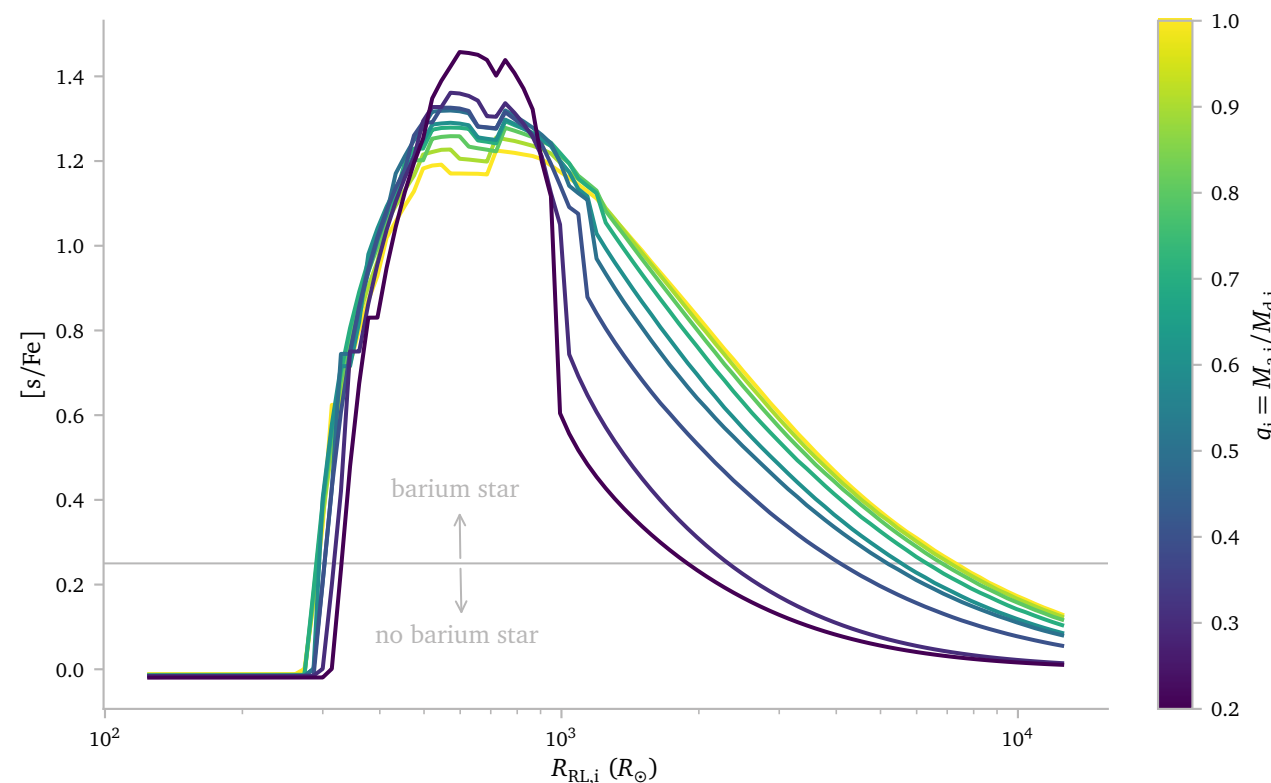


Clearly, when plotting the radius as a fraction of the envelope (and also, but less clearly, just as the envelope mass), we can see that the last thermal pulses for the various models happen at different remaining envelope fractions.

The $1.6 M_{\odot}$ model has relatively a lot more envelope mass remaining when the last pulse happens than the $2.8 M_{\odot}$ model. I think it then makes sense why we see the sharp drop when the models miss RLOF and why the height of this drop can depend quite strongly on the mass. It is really an effect of how much mass is remaining during the last TP.

We can also change the initial mass ratio q . This should have some interesting effects. In particular, the [Saladino et al. \(2018, 2019\)](#) wind prescription is dependent on q , q shifts the Roche lobe radius in a non-trivial way. A smaller q will mean that R_{RL}/a is larger, which means for the same initial R_{RL} , the initial separation a_i is larger.

Lastly, the RLOF mass transfer is depends on the initial mass ratio due to the fraction $\Delta M_{\text{acc}}/M_{\text{acc}}$. In this simple case, ΔM_{acc} is constant, while $M_{\text{acc}} = q \cdot M_{\text{TPGB},i}$. So for smaller q , we expect larger abundances during RLOF.

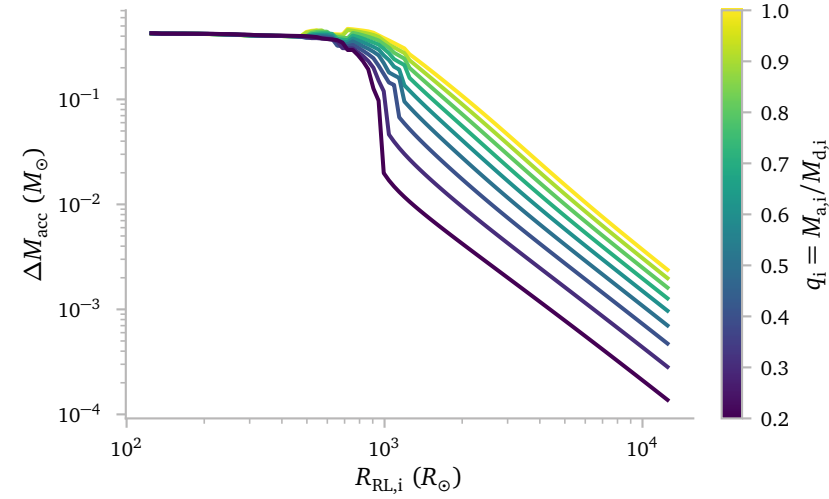


We indeed see that the dependence on q is quite important. In this simple case, a smaller q will lead to *higher* abundances during RLOF, but *lower* abundances during the wind mass loss regime.

In reality, the effect is probably more nuanced. I would not at all be surprised if there is some effect of q on the mass transfer efficiency. Since models with lower q tend to experience much more extreme RLOF, with higher mass loss rates, it would not surprise me if more of this mass is lost, which would counteract the effect of higher abundances due to the lower accretor mass.

We also see that the discontinuity between the RLOF and the wind regime is much more pronounced for the lower q models. I think this is probably because the mass transfer efficiency of wind is lowered with lower q , while this is not the case during RLOF.

Let me explicitly test this. Below, I show the amount of accreted mass for the different q models.



We can indeed clearly see that the models with lower q experience much less wind accretion.

There also exists a nice powerlaw between the accreted material due to wind and the initial Roche lobe radius (or initial binary separation).

References

- Asplund, M., Grevesse, N., Sauval, A. J., & Scott, P. (2009) *The Chemical Composition of the Sun*. [ARA&A47](#), 481.
- Karakas, A. I. & Lugaro, M. (2016) *Stellar Yields from Metal-rich Asymptotic Giant Branch Models*. [ApJ825](#), 26.
- Saladino, M. I., Pols, O. R., & Abate, C. (2019) *Slowly, slowly in the wind. 3D hydrodynamical simulations of wind mass transfer and angular-momentum loss in AGB binary systems*. [A&A626](#), A68.
- Saladino, M. I., Pols, O. R., van der Helm, E., Pelupessy, I., & Portegies Zwart, S. (2018) *Gone with the wind: the impact of wind mass transfer on the orbital evolution of AGB binary systems*. [A&A618](#), A50.